Appendix B: Preprocessing excerpt *preprocessing/pingstation_preprocess.py*

```
from datetime import datetime, timezone


class PingstationPreprocessor:
    def _first_present(self, raw_data, *keys, default=None):
        for key in keys:
            if key in raw_data and raw_data[key] not in (None, ""):
                return raw_data[key]
        return default

    def _to_float(self, value, default=0.0):
        try:
            return float(value)
        except (TypeError, ValueError):
            return default

    def clean_altitude_mm(self, altitude):
        """Normalizes altitude to millimeters."""
        if altitude is None:
            return 0

        if isinstance(altitude, str):
            text = altitude.strip().lower().replace(",", "")
            if not text:
                return 0

            if "ft" in text:
                feet = self._to_float(text.replace("ft", "").strip())
                meters = feet * 0.3048
                return int(meters * 1000)

            if text.endswith("m"):
                meters = self._to_float(text.replace("m", "").strip())
                return int(meters * 1000)

            numeric = self._to_float(text)
            return int(numeric if numeric > 100000 else numeric * 1000)

        numeric = self._to_float(altitude)

        # Pingstation-style values are often already in mm and usually larger than 100k.
        if numeric > 100000:
            return int(numeric)

        # Otherwise treat as meters.
        return int(numeric * 1000)

    def clean_heading_de2(self, heading):
        """Normalizes heading to degrees*100."""
        if heading is None:
            return 0

        if isinstance(heading, str):
            heading = heading.strip().replace("deg", "").replace("°", "")

        value = self._to_float(heading)

        if value <= 360:
            return int(value * 100)

```